

Sander Beckers

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Research areas. Causation and counterfactuals; causal models and their semantics; explanation, harm and responsibility in AI; logic and formal epistemology; philosophy of science.

EMPLOYMENT

2025–present	Postdoctoral researcher , Department of Statistical Science, University College London. UKRI Causality in Healthcare AI hub (CHAI). Supervisor: Ricardo Silva.
2024–2025	Postdoctoral researcher , Department of Computer Science, Cornell University. Supervisor: Joseph Y. Halpern.
2023–2024	Postdoctoral researcher , Institute for Logic, Language and Computation, University of Amsterdam. Supervisor: Robert van Rooij.
2021–2023	Postdoctoral researcher , Cluster of Excellence <i>Machine Learning for Science</i> , University of Tübingen. Supervisor: Konstantin Genin.
2019–2021	Humboldt Research Fellow , Munich Center for Mathematical Philosophy, LMU Munich. Supervisor: Stephan Hartmann.
2018–2019	Postdoctoral scholar , Department of Philosophy and Religious Studies, Utrecht University. Supervisor: Jan Broersen.
2016–2017	BAEF Postdoctoral Fellow , Department of Computer Science, Cornell University. Supervisor: Joseph Y. Halpern.
2010–2011	Researcher , EAVISE group, University of Leuven. Parallel SAT solving on GPUs (project <i>S.O.S. OpenCL</i>).
2008–2009	Actuarial intern , Aon Benfield Belgium and BNP Paribas Belgium.

EDUCATION

2012–2016	PhD in Computer Science , University of Leuven. <i>Actual Causation: Definitions and Principles</i> . Supervisors: Hendrik Blockeel, Joost Vennekens.
2008–2011	Master in Mathematics (distinction), University of Leuven. <i>Deductive Reasoning in Guarded FO(ID)</i> . Supervisor: Marc Denecker. Preceded by a bridging programme in mathematics (2008).
2004–2006	Master in Philosophy (distinction), University of Leuven. <i>Wittgenstein en het Ethische</i> . Supervisor: Arnold Burms.
2002–2004	Bachelor in Philosophy (distinction), University of Antwerp.

GRANTS, FELLOWSHIPS AND AWARDS

2026–2028	Australian Research Council Discovery Project , partner investigator. <i>An Interventionist Approach to Explainable Artificial Intelligence</i> . Lead: Samuel Baron, University of Melbourne.
2025	Top Reviewer , UAI 2025.
2023	Best Paper Award , CLeaR 2023, for <i>Backtracking Counterfactuals</i> .
2020	NIAS-Lorentz Theme Group fellowship, Netherlands Institute for Advanced Study (explainable AI in medicine).
2019–2021	Humboldt Research Fellowship for Postdoctoral Researchers.
2016–2017	Research Fellowship , Belgian American Educational Foundation.
2012–2015	PhD Grant for Fundamental Research , FWO.

PUBLICATIONS

1. Beckers, S. (2026). Large Language Models as Nondeterministic Causal Models. *KR 2026*.
2. Lally, J., Kazemi, M., Paoletti, N., Watson, D., Beckers, S. (2026). Robust Counterfactual Policy Optimisation via Nondeterministic Causal Models. *UAI 2026 workshop*.
3. Beckers, S. (2025). What Does It Take to Make a Difference? A Reply to Andreas and Günther. *Journal of Philosophy* 122(4), 177–189.
4. Beckers, S. (2025). Nondeterministic Causal Models. *CLear 2025*, PMLR 275, 1532–1554.
5. Beckers, S. (2025). Actual Causation and Nondeterministic Causal Models. *CLear 2025*, PMLR 275, 514–532.
6. Beckers, S., Chockler, H., Halpern, J.Y. (2024). A Causal Analysis of Harm. *Minds and Machines* 34(34).
7. Belle, V., Chockler, H., Vallor, S., Varshney, K.R., Vennekens, J., Beckers, S. (2024). Trustworthiness and Responsibility in AI: Causality, Learning and Verification (Dagstuhl Seminar 24121). *Dagstuhl Reports*.
8. Beckers, S. (2023). Moral Responsibility for AI Systems. *NeurIPS 2023*.
9. Beckers, S., Chockler, H., Halpern, J.Y. (2023). Quantifying Harm. *IJCAI 2023*, 363–371.
10. Beckers, S., Halpern, J.Y., Hitchcock, C. (2023). Causal Models with Constraints. *CLear 2023*, PMLR 213, 866–879.
11. von Kügelgen, J., Mohamed, A., Beckers, S. (2023). Backtracking Counterfactuals. *CLear 2023*, PMLR 213, 177–196. **Best Paper Award**.
12. Beckers, S., Chockler, H., Halpern, J.Y. (2022). A Causal Analysis of Harm. *NeurIPS 2022*.
13. Beckers, S. (2022). Causal Explanations and XAI. *CLear 2022*, PMLR 177, 90–109.
14. Beckers, S. (2021). Causal Sufficiency and Actual Causation. *Journal of Philosophical Logic* 50, 1341–1374.
15. Beckers, S. (2021). The Counterfactual NESS Definition of Causation. *AAAI 2021*, 6210–6217.
16. Beckers, S. (2021). Equivalent Causal Models. *AAAI 2021*, 6202–6209.
17. Beckers, S., Eberhardt, F., Halpern, J.Y. (2019). Approximate Causal Abstraction. *UAI 2019*.
18. Beckers, S., Halpern, J.Y. (2019). Abstracting Causal Models. *AAAI 2019*, 2678–2685.
19. Beckers, S. (2018). AAAI: An Argument Against Artificial Intelligence. In V.C. Müller (ed.), *Philosophy and Theory of Artificial Intelligence 2017*, Springer, 235–247.
20. Beckers, S., Vennekens, J. (2018). A Principled Approach to Defining Actual Causation. *Synthese* 195(2), 835–862.
21. Beckers, S., Vennekens, J. (2017). The Transitivity and Asymmetry of Actual Causation. *Ergo* 4(1), 1–27.
22. Beckers, S., Vennekens, J. (2016). A General Framework for Defining and Extending Actual Causation Using CP-logic. *International Journal of Approximate Reasoning* 77, 105–126.
23. Beckers, S., Vennekens, J. (2015). Towards a General Definition of Actual Causation Using CP-logic. *PLP@ICLP 2015*, CEUR 1413, 19–38.
24. Beckers, S., Vennekens, J. (2015). Combining Probabilistic, Normative and Causal Reasoning in CP-logic. *Commonsense 2015*, 32–38.
25. Beckers, S., Vennekens, J. (2012). Counterfactual Dependency and Actual Causation in CP-logic and Structural Models: A Comparison. *STAIRS 2012*, FAIA 241, 35–46.
26. Beckers, S., et al. (2012). Parallel Hybrid SAT Solving Using OpenCL. *BNAIC 2012*, 11–18.
27. Beckers, S., et al. (2011). Parallel SAT-Solving with OpenCL. *IADIS Applied Computing 2011*, 435–441.

SUPERVISION

2026–2028	PhD co-supervisor (with Hana Chockler): Başak Kocaoğlu, PhD in Informatics, King's College London.
2025	PhD committee, final reader: Luna De Souter, Philosophy of Science, University of Bergen.
2023	PhD committee, substitute member: Dean McHugh, Logic, University of Amsterdam.
2018–2024	Three master's theses supervised (Utrecht, Amsterdam).

TEACHING

- 2023–2024 **University of Amsterdam.** *Philosophy & AI*, BSc Artificial Intelligence, lecturer, two years (mandatory course, 130 students). *Philosophical Logic*, Master of Logic, lecturer (2023). *Actual Causation*, Master of Logic seminar (2024).
- 2018–2019 **Utrecht University.** *Introduction to Statistics*, BSc Artificial Intelligence, lecturer (2018). *Philosophy of AI*, MSc Artificial Intelligence, tutorials and guest lectures. Tutorials in *Logic for Philosophers*, *Epistemology and Philosophy of Science*, and *Philosophical Reflection on Scientific Models of Man*, BA Philosophy.
- 2010–2015 **University of Leuven.** Exercise classes in *Introduction to Programming* in C (2010–2013), Java (2012–2013) and Python (2014–2015), BSc Engineering; *Introduction to the IDP Knowledge Base System* (2015).

INVITED TALKS

Forty-plus invited talks since 2012, fourteen of them in 2025–2026. Selected:

- 2026 **Keynote**, CREST workshop (Causation, Responsibility and Explanations in Science and Technology), FLoC 2026, Lisbon · **Invited panelist**, American Causal Inference Conference, Salt Lake City · **Invited panelist** with Philip Dawid and Nancy Cartwright, EuroCIM 2026, Oxford · Foundations of Causal Inference programme, **Isaac Newton Institute**, Cambridge · Department of Statistics, LSE · the Choice Group, LSE · NLP seminar, King's College London · Utrecht University
- 2025 **Keynote**, UAI 2025 workshop on Causal Abstractions and Representations, Rio de Janeiro · **Keynote**, closing workshop of the Bias, Risk and Opacity in AI project, University of Milan · Causal AI Lab, Columbia University · Experimental Psychology, UCL · Interdisciplinary Symposium on AI & Society, Warwick · University of Exeter · University of Bergen · CHAI seminar
- 2024 MCMP colloquium, LMU Munich · Dagstuhl Seminar on Trustworthiness and Responsibility in AI · Causal Abstraction, Reasoning and Extrapolation seminar
- 2023 PALLMYR, UCL · Kyoto Causation Workshop · Causality in Practice, Institut Pascal, Paris · Leibniz AI Lab symposium on AI, Causality and Personalised Medicine · Amsterdam Causality Meeting · SIKS course on Human-Centered AI · SAP Inspiration Sessions · Workshop on Causation and Modality, Amsterdam
- 2012–2022 Workshop on Causal Representation Learning, UAI 2022 · Online Causal Inference Seminar (2022) · Monash University roundtable on AI and legal institutions (2021) · Logic Seminar, Milan (2021) · CUNY Logic and Metaphysics Workshop (2018) · Center for Human-Compatible AI, UC Berkeley (2017) · MIT Linguistics and Philosophy (2015) · earlier talks at Nijmegen, Düsseldorf, Tilburg, Leuven, LMU Munich and INRIA Grenoble

RESEARCH VISITS

- 2022 History and Philosophy of Science, University of Pittsburgh. Host: James Woodward.
- 2020 NIAS, Amsterdam. Lorentz theme group on explainable AI in medicine.
- 2019 Humanities and Social Sciences, Caltech. Host: Frederick Eberhardt.
- 2017 Institute of Philosophy, Leipzig University. Host: James Conant.

PROFESSIONAL SERVICE

- Organisation Workshop on Causality for Ethics and Society, LMU Munich, 2023 (with Naftali Weinberger). Co-author of the Dagstuhl Seminar 24121 report.
- Grant reviewing ELLIS PhD programme; French National Research Agency (ANR); Netherlands Institute for Advanced Study (×3).

Programme committees	AAAI (2017, 2020–2024), NeurIPS (2023–2025), UAI (2024–2026), ICLR (2026), CLeaR (2024, 2025), AIES (2024), FAccT (2023), AISTATS (2022), IJCAI (2020); workshops at AAAI, NeurIPS, ICML and AAMAS.
Journal refereeing	<i>Journal of Machine Learning Research, Journal of the American Statistical Association, IEEE Transactions on AI, Journal of Causal Inference, Journal of Logic and Computation, Review of Symbolic Logic, Mind, The Philosophical Review, Journal of Philosophy, Philosophy of Science, British Journal for the Philosophy of Science, Synthese, Erkenntnis, Minds and Machines, Philosophical Studies, Journal of Philosophical Logic, Analysis, Ethics and Information Technology, European Journal for Philosophy of Science, AI & Society, and others.</i>

LANGUAGES

Dutch (native), English (fluent), French (fluent), German (intermediate).